JINGQING LIU

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EDUCATION

UNIVERSITY OF DELAWARE
B.S. in Computer Science
Cybersecurity Concentration
Expected in Jun 2024
Cum. GPA: 3.96/4.0

Online Research Seminar
Topic: Cryptography, Cybersecurity, and Blockchains
Advisor: Dr. Vipul Goyal (Carnegie Mellon University)

LINK

https://github.com/Jingqing-Liu

TA EXPERIENCE

TEACHING ASSISTANT
University of Delaware, Mathematics Dept., Computer Science Dept.

ACTIVITIES

UDCSSA MID-AUTUMN FESTIVAL PARTY
09/2021
※Delivered a live broadcast; Handled unexpected situations

IT DEPARTMENT DIRECTOR, UDCSSA GAME CONTEST
08/2021
※Remotely controlled the live content to ensure error-free live streaming

PEER NOTE TAKER, DELTA ALPHA PI
09/2020-12/2020
※Shared study notes with disabled students

PROJECT

Parallel Computing N-Body Simulator
Spring 2023 | U DELAWARE
※Developed a program to track/compute over time the movement and speed of the objects in the solar system
※Ranked among the Top 3 in the class of 45

SKILLS

PROGRAMMING
Javascript, Typescript, Python, PyQt, Java, C++, C, R

INTERESTS
Piano (Gr. 10), Taekwondo (Black Belt)

MEMBERSHIPS

※Golden Key International Honor Society Membership
※Tau Sigma National Honor Society Membership
※Phi Kappa Phi Membership

RESEARCH EXPERIENCES

X-Lab of Dr. Xing Gao | Advisor: Xing Gao (Assistant Professor, U Delaware)
06/2023- Present | University of Delaware

Project I: Research and Testing of API Calls in WeChat
※Comprehensively examined WeChat's API system, focusing on its technical operation principles and associated security vulnerabilities
※ Found the variability in API implementation across different operating systems, unauthorized exposure of system resources and discrepancies in permission settings, the threat of sub-window spoofing, and other security risks

Project II: Invisible Threats in the Met-Averse: Roblox's Security Vulnerabilities
※Explored how Roblox's infrastructure circumvents firewall restrictions; Found that Roblox can be manipulated by local malware to launch connections
※ Found that Roblox's distributed computing resources can be used for resource-intensive tasks such as neural network training

Verification of Account Balances in the Bitcoin System Using Merkle Trees
2023 Summer
Advisor: Vipul Goyal (Professor, Carnegie Mellon University)
※Delved into the complex structure of the Bitcoin protocol, focusing on block architecture, node categorization, and transaction validation mechanisms
※Proposed an innovative solution that integrates the Merkle tree root of the UTXO set into the unused portion of the coinbase transaction input field, to address the challenges of verifying light node account balances
※Proposed a solution that allows light nodes to use Merkle proofs when verifying their account balances, thus improving accuracy and trust; Ensured the alignment with soft fork standards, incentivizing node upgrades while maintaining the integrity of the network

GUI Development for Water Network Assessment and Simulation System
09/2022- 06/2023 | University of Delaware
Advisor: Sina Naeimi Dafchahi (Doctoral Candidate / Graduate Research Assistant)
※ Developed a Graphic User Interface based on PyQT according to the research needs and the provided framework; Adopted a multiple window format
※ Improved the existing user interface based on Python's own GUI module, Tkinter, by enhancing its outlook, usability, and scalability

A Design of Router Firewall | Advisor: Rui Zhang (Associate Prof., U Delaware)
06/2022-08/2022 | Newark
※ Conducted a survey with nearly 1000 respondents; Found that user errors caused most security breaches
※ Designed a router-based firewall system that eliminates human operation errors and thus better protects users' computers from Viruses and illegal websites

PUBLICATION

Jingqing Liu (First Author); ***Enhancing Network Security Through Router-Based Firewalls: An Investigation into Design, Effectiveness, and Human Factors***; accepted by ***2023 5th International Conference on Computer Science and Intelligent Communication*** (CSIC 2023); will be published in ***Highlights in Science, Engineering and Technology***, ISSN: 2791-0210

INTERNSHIPS

BEIJING EETRUST TECH CO, LTD. | INTERN SOFTWARE ENGINEER
05/2022- 08/2022 | Beijing, China
※Built logging systems for banks and insurance companies, including the service console and terminal clients; Improved system security
※Maintained both the service console and terminal login clients

CHINA CONSTRUCTION BANK | INTERN HUMAN RESOURCE DEPT.
06/2021- 08/2021 | Shanghai, China
※Analyzed data to find out the reasons for staff turnover

HONORS

2023	U DELAWARE	Summer Scholar and Scholarship
2020/2023	U DELAWARE	Dean's List (All Semesters)
2022	U DELAWARE	Summer Scholar and Scholarship
2022	U DELAWARE	CIS Undergrad Honorable Mention
2020	U DELAWARE	1st Place, "Best in Show" Category for Product Design